

Project 2, parts 4 and 5: RNA-seq analysis

2025-05-08

Project 2 parts 4 and 5 overall: 75 points

Include any code that you used to solve the question as well as the answers for all below questions. In parts 4 and 5 from project 2, you will explore how gene expression differs between electric fish tissues. You will learn how to analyze RNA-seq count data: identify differentially expressed genes, distinguish samples, and make biological interpretations. Additionally, you will visualize findings and communicate them effectively. **Note: Please submit the Rmd and the knit HTML file for this assignment**

Dataset reminder:

In this project, you are be analyzing the count data and identifying differentially expressed genes between skeletal muscle and electric organ tissue. As a reminder, these datasets originated from two different electric fish studies (PRJNA1005245 and PRJNA1005244). It may help to consider how these papers analyzed the data: the methods for the PRJNA1005244 dataset are published and the methods for the PRJNA1005245 dataset are written in the third chapter of a thesis. These studies are both using the genus *Campylomormyrus*. **For this assignment, you may not need to analyze all of the data to address the hypothesis.**

Resources:

There are several resources that you may use to complete this assignment: 1) A primary resource: DESeq2 workflow. 2) DESeq2's additional guide 3) Harvard Bioinformatics Core training.

You are expected to use DESeq2 for this assignment. If you decide to use an additional package, please list the package name, how you found it, and why you chose to use it. We will do our best to help you navigate other packages, but we cannot guarantee support for other packages.

Project 2 part 4: Hypothesis testing and exploration

0. In part 4, your goal is to investigate differences in gene expression between the electric organ and skeletal muscle. Write the null hypothesis for this goal below.

[There is no difference in gene expression between the electric organ and skeletal tissue.]

1. Load your data into DESeq2

- Your data is located on Talapas: /projects/bgmp/shared/Bi623/Project2/counts.tar
- **Think carefully about your design formula and explain your chosen formula below**
- Note: you can follow the Bear_analysis rmd from class as a template or can also reference other resources online, such as this guide from Duke to load your data.

```
library(edgeR)
```

```
## Loading required package: limma
```

```
library(limma)
library(Glimma)
library(ggplot2)
library(RColorBrewer)
library(DESeq2)
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
## Loading required package: generics
```

```
##
```

```
## Attaching package: 'generics'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##      setequal, union
```

```
##
```

```
## Attaching package: 'BiocGenerics'
```

```
## The following object is masked from 'package:limma':
```

```
##
```

```
##      plotMA
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      IQR, mad, sd, var, xtabs
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##      colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##      get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##      order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##      rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##      unsplit, which.max, which.min
```

```
##
```

```
## Attaching package: 'S4Vectors'
```

```

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

## Loading required package: IRanges

##
## Attaching package: 'IRanges'

## The following object is masked from 'package:grDevices':
##
##     windows

## Loading required package: GenomicRanges

## Loading required package: Seqinfo

## Loading required package: SummarizedExperiment

## Loading required package: MatrixGenerics

## Loading required package: matrixStats

##
## Attaching package: 'MatrixGenerics'

## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars

## Loading required package: Biobase

```

```

## Welcome to Bioconductor
##
##   Vignettes contain introductory material; view with
##   'browseVignettes()'. To cite Bioconductor, see
##   'citation("Biobase")', and for packages 'citation("pkgname)".

##
## Attaching package: 'Biobase'

## The following object is masked from 'package:MatrixGenerics':
##
##   rowMedians

## The following objects are masked from 'package:matrixStats':
##
##   anyMissing, rowMedians

library(apeglm)
library(tools)
library(qvalue)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:Biobase':
##
##   combine

## The following object is masked from 'package:matrixStats':
##
##   count

## The following objects are masked from 'package:GenomicRanges':
##
##   intersect, setdiff, union

## The following object is masked from 'package:Seqinfo':
##
##   intersect

## The following objects are masked from 'package:IRanges':
##
##   collapse, desc, intersect, setdiff, slice, union

## The following objects are masked from 'package:S4Vectors':
##
##   first, intersect, rename, setdiff, setequal, union

## The following objects are masked from 'package:BiocGenerics':
##
##   combine, intersect, setdiff, setequal, union

```

```
## The following object is masked from 'package:generics':  
##  
##     explain
```

```
## The following objects are masked from 'package:stats':  
##  
##     filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##     intersect, setdiff, setequal, union
```

```
library(data.table)
```

```
##  
## Attaching package: 'data.table'
```

```
## The following objects are masked from 'package:dplyr':  
##  
##     between, first, last
```

```
## The following object is masked from 'package:SummarizedExperiment':  
##  
##     shift
```

```
## The following object is masked from 'package:GenomicRanges':  
##  
##     shift
```

```
## The following object is masked from 'package:IRanges':  
##  
##     shift
```

```
## The following objects are masked from 'package:S4Vectors':  
##  
##     first, second
```

```
## The following object is masked from 'package:base':  
##  
##     %notin%
```

```
library(pheatmap)  
library(PoiClaClu)  
library(glmPCA)  
library(tibble)  
library(ashr)
```

```
##  
## Attaching package: 'ashr'
```

```
## The following object is masked from 'package:generics':
##
##      prune
```

#Include your commands, remember that you might not need to load all of the count files for this hypothesis

Load metadata

```
metadata = read.csv("./Project2_Part4/Project2_RNAseq_metadata_campy.csv")
```

[My design formula is ~SpeciesSmaller + Tissue + Species_Smaller:Tissue. I am interested in seeing how the gene expression varies between species while also controlling for tissue. I have also included in my design formula ("SpeciesSmaller:Tissue") an assumption that the expression effect of species is impacted by tissue since the tissue inherently varies by the individual it was sourced from. I am working in tandem with Zach on this as he is doing this exact design formula without this extra ':' statement so we can compare our results.]

2. Filter the dataset to remove genes that are lowly expressed

- There are multiple ways to filter RNA-Seq datasets. You can either use cpm as a filter as we did in the Bear_analysis or use raw counts completed in the DESeq2 vignette from the above resources.
- Describe why you used your filtering strategy

#Filter out all the baby fish since they don't have skeletal tissue

```
metadata = subset(metadata, metadata$AgeSize == "adult")
```

```
Campy = DESeqDataSetFromHTSeqCount(metadata,directory="Project2_Part4/counts_tar",design=~SpeciesSmaller
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

Obtain CPMs

```
myCPM <- cpm(Campy)
```

Which values in myCPM are greater than 0.5?

```
thresh <- myCPM > 0.5
```

we would like to keep genes that have at least 2 TRUES in each row of thresh

```
keep <- rowSums(thresh) >= 2
```

Subset the rows of count data to keep the more highly expressed genes

```
Campy <- Campy[keep,]
```

[I chose the counts per million approach because, similar to the study, we removed low count transcripts and only kept transcripts whose expression was above in cpm threshold in at least two separate samples. This is important to ensure that every highly expressed gene has a "replicate" in another sample to filter out outliers unique to specific samples. Unlike the study I didn't pass in a specific count threshold (they passed in 10), I used a cpm ratio of 0.5 instead per the Bear study example. Using a cpm ratio is a bit more flexible than passing in a hard count so you can eliminate low count transcripts relative to the size of the transcriptome.]

3. Normalize the dataset and run differential expression analysis

- You may choose to normalize and run differential expression separately as done in the Bear_analysis, or in 1 command as done in the DESeq workflow 5.1.

```
# Normalize the data
Campy = DESeq(Campy)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```
## 1 rows did not converge in beta, labelled in mcols(object)$betaConv. Use larger maxit argument with
```

4. Normalize the data to use for dimensionality reduction

- Use the rlog transformation method (follow along with the workflow, section 4.2)
- Describe why the rlog transformation is performed

```
Campy_rlog = rlog(Campy, blind = FALSE)
```

```
## rlog() may take a few minutes with 30 or more samples,
```

```
## vst() is a much faster transformation
```

[We used the rlog function to normalize the data in preparation to conduct and plot a principal component analysis. Without log normalization, the genes with the highest counts (and thus the highest expression) would be over-represented in these analyses and would mask any findings from the lesser expressed genes.]

5. Identify distances between samples using sample distance heatmaps

- Generate a Euclidean distance heatmap and dendrogram using your rlog transformed data (follow workflow section 4.3)
- Create a Poisson distance heatmap with dendrogram using DESeq data (follow workflow section 4.3)
- Describe what each of these heatmaps are measuring
- Look at the sample groupings for each heatmap+dendrogram. Do you notice any potential individual outliers? Do both plots agree?

Note: You may need to install the packages “pheatmap” and “RColorBrewer” to follow along. If R tells you to restart your session listen to it.

```
#Include your commands for generating the Euclidean distance heatmap and dendrogram with the rlog data
campyDists = dist(t(assay(Campy_rlog)))
```

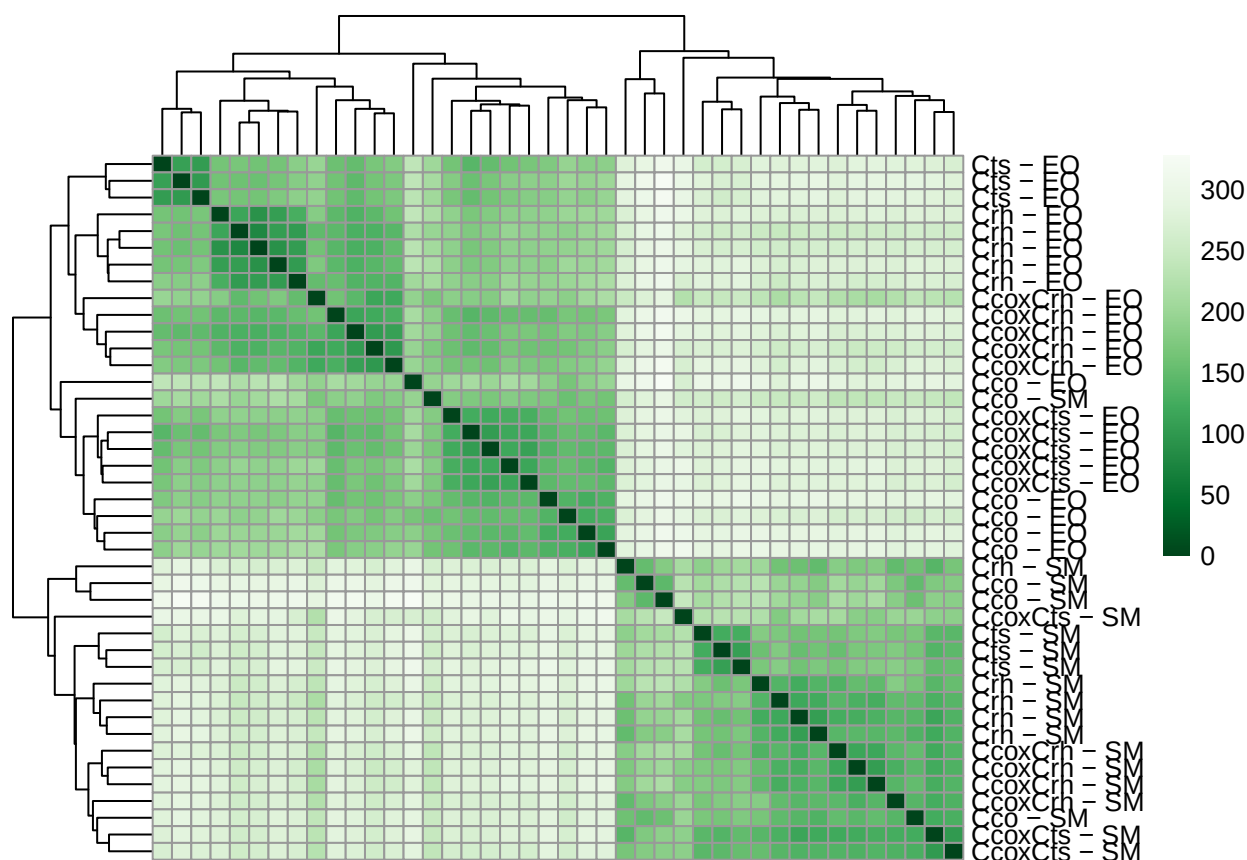
```
campyDistMatrix = as.matrix(campyDists)
```

```
rownames(campyDistMatrix) = paste(Campy_rlog$SpeciesSmaller, Campy_rlog$Tissue, sep = " - " )
```

```
colnames(campyDistMatrix) = NULL
```

```
colors <- colorRampPalette(rev(brewer.pal(9, "Greens")) )(255)
```

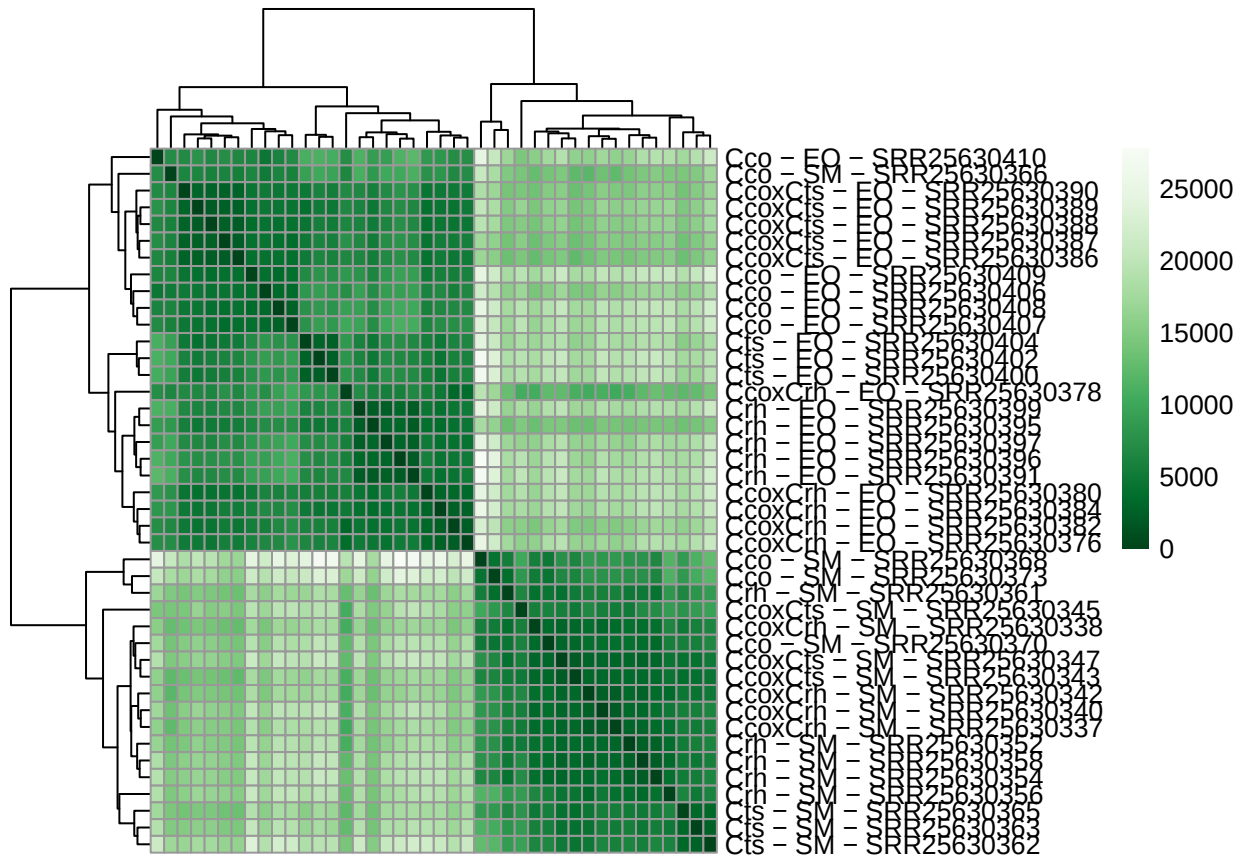
```
pheatmap(campyDistMatrix,
          clustering_distance_rows = campyDists,
          clustering_distance_cols = campyDists,
          col = colors)
```



#Include your commands for generating the Poisson distance heatmap with dendrogram using your original

```
campyPois = PoissonDistance(t(counts(Campy)))

campyPoisDistMatrix = as.matrix(campyPois$dd)
rownames(campyPoisDistMatrix) = paste(Campy$SpeciesSmaller, Campy$Tissue, Campy$SRRfile, sep=" - ")
colnames(campyPoisDistMatrix) = NULL
pheatmap(campyPoisDistMatrix,
          clustering_distance_rows = campyPois$dd,
          clustering_distance_cols = campyPois$dd,
          col = colors)
```

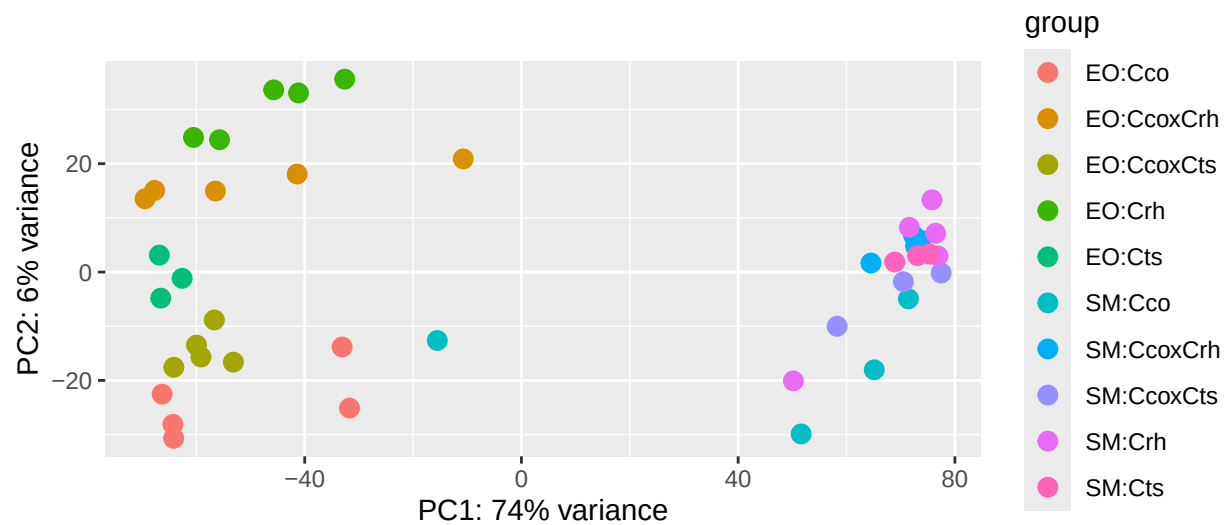
[Both of these heatmaps are visualizing the clusters of expression differences in a matrix format. That's why there is a diagonal line through it- it represents the expression difference between identical samples, thus why it is 0. The plots both show that the EO and SM tissues are clustering together with the exception of one sample (Cco-SM) that is clustering with the EO samples in both plots. Of note, the Poisson plot displays more noise than the Euclidean plot.]

6. Examine similarity in sample global gene expression using PCA, Generalized PCA, and MDS plots.

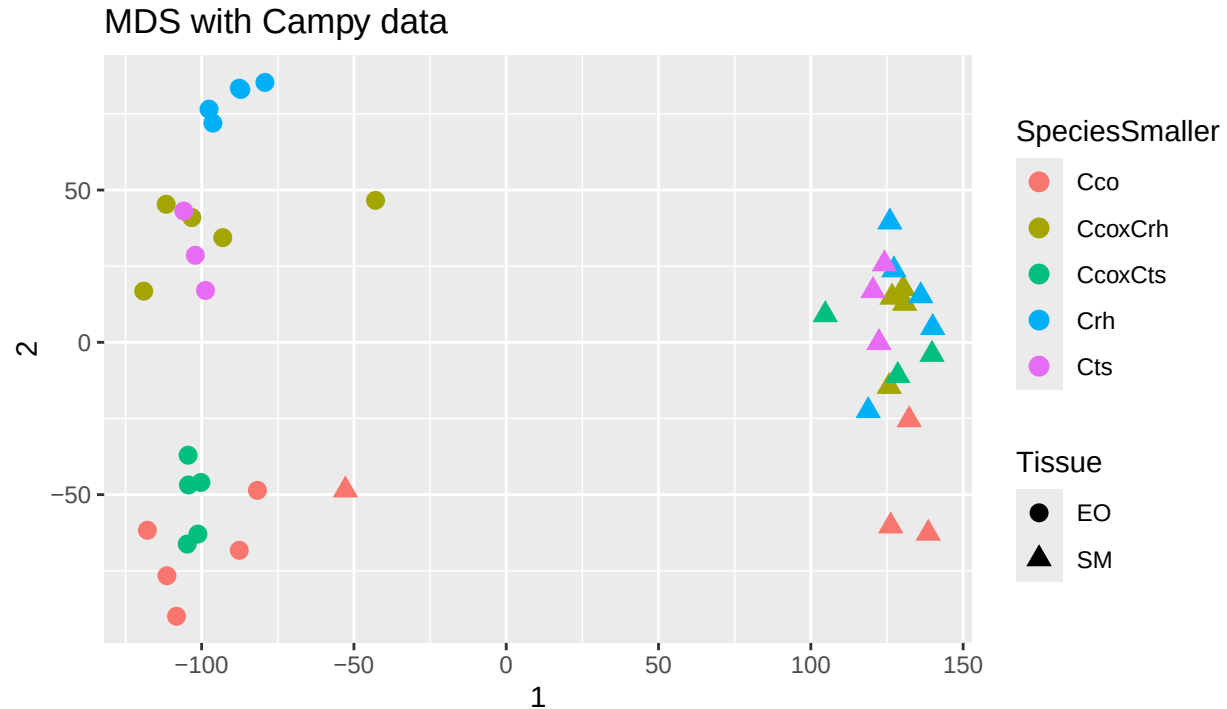
- Generate PCA, Generalized PCA, and MDS plots (following workflow section 4.4-4.6)
- Modify the color and shape of points to reflect Tissue and Species of samples
- Briefly describe each approach and their findings.

```
#PCA
plotPCA(Campy_rlog, intgroup = c("Tissue", "SpeciesSmaller"))
```

```
## using ntop=500 top features by variance
```



```
# Generalized PCA
gpca <- glmpca(counts(Campy), L=2)
gpca.dat <- gpca$factores
gpca.dat$SpeciesSmaller <- Campy$SpeciesSmaller
gpca.dat$Tissue <- Campy$Tissue
ggplot(gpca.dat, aes(x = dim1, y = dim2, color = SpeciesSmaller, shape = Tissue)) +
  geom_point(size = 3) + coord_fixed() + ggtitle("Generalized PCA of Species and Tissue Type")
```

[All of the above approaches cluster very clearly between tissue types with the exception of the deviant skeletal muscle sample. However, they don't all differ between species equally. The MDS plot (which is the plot the researchers ended up putting in their paper) provides the clearest clustering between species followed by the PCA which mildly follows a similar clustering pattern and then the generalized PCA doesn't visualize any clustering between species. The regular PCA assumes your data follows a normal distribution while the generalized PCA does not assume this whereas the MDS approach specifically is designed for examining a distance matrix to compute pair-wise distances. The distance matrix we calculated for the heatmaps above is perfect for an MDS plot.]

7. Use your dimensionality reduction plots, evaluate sample quality and complete quality adjustments.

- Based on your dimensionality reduction plots, do you have concerns about any samples? Explain.
- If so, remove them now, repeat steps of loading in data, filtering, normalization, and running the differential expression analysis.

[There is one skeletal muscle sample that consistently clustered with the electric organ samples in every dimensionality plot, so I would want to remove this sample from my metadata. See code below for this and recomputing all the calculations without it.]

```
# Redo calculations without deviant skeletal muscle sample from SRR25630366 dataset
```

```
metadata = filter(metadata, metadata$SRRfile != "SRR25630366")
```

```
Campy = DESeqDataSetFromHTSeqCount(metadata, directory="Project2_Part4/counts_tar", design=~SpeciesSmaller
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

```
# Obtain CPMs
myCPM <- cpm(Campy)
# Which values in myCPM are greater than 0.5?
thresh <- myCPM > 0.5
# we would like to keep genes that have at least 2 TRUES in each row of thresh
keep <- rowSums(thresh) >= 2
# Subset the rows of count data to keep the more highly expressed genes
Campy <- Campy[keep,]

# Normalize
Campy = DESeq(Campy)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```
# Normalize for dimension reduction
Campy_rlog = rlog(Campy, blind = FALSE)
```

```
## rlog() may take a few minutes with 30 or more samples,
## vst() is a much faster transformation
```

8. Identify genes that are differentially expressed between SM and EO tissues in EVERY species comparison.

- Generate a results table using log fold shrinkage and proper contrasts
- Refer to this reference for guidance on extracting contrasts
- When setting contrasts, identify the baseline variables used in the intercept to help determine proper contrasts
- Filter DEGs to contain only candidates with a $|\log_2\text{FoldChange}| > 1$ and adjusted value $< .05$
- Join DEG data sets for every species, remove any DEG not found in every species, and calculate the average log2FC and average adjusted p-value across all species
- Visualize your cleaned DEG dataset with a volcano plot.
 - Use the average log2FC and adjusted p-values for plotting
 - Make sure your plot indicates the meaning of negative and positive logfold changes

```
## Generation of result tables
```

```
# CCO
```

```
CCO = data.frame(lfcShrink(dds = Campy, coef = "Tissue_SM_vs_E0"), type = "ashr") %>%  
  rownames_to_column(var = "genes") %>%  
  filter(padj < 0.05) %>%  
  filter(abs(log2FoldChange) > 1) %>%  
  rename("CcoLog2FoldChange" = "log2FoldChange", "Ccopadj" = "padj") %>%  
  select(genes, CcoLog2FoldChange, Ccopadj)
```

```
## using 'apeglm' for LFC shrinkage. If used in published research, please cite:  
##   Zhu, A., Ibrahim, J.G., Love, M.I. (2018) Heavy-tailed prior distributions for  
##   sequence count data: removing the noise and preserving large differences.  
##   Bioinformatics. https://doi.org/10.1093/bioinformatics/bty895
```

```
# CCO x CRH
```

```
CCOxCRH = data.frame(lfcShrink(dds = Campy, contrast = list(c("SpeciesSmallerCcoxCrh.TissueSM", "Tissue_
```



```
  rownames_to_column(var = "genes") %>%  
  filter(padj < 0.05) %>%  
  filter(abs(log2FoldChange) > 1) %>%  
  rename("CcoxCrhLog2FoldChange" = "log2FoldChange", "CcoxCrhpadj" = "padj") %>%  
  select(genes, CcoxCrhLog2FoldChange, CcoxCrhpadj)
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:  
##   Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.  
##   https://doi.org/10.1093/biostatistics/kxw041
```

```
# CCO x CTS
```

```
CCOxCTS = data.frame(lfcShrink(dds = Campy, contrast = list(c("SpeciesSmallerCcoxCts.TissueSM", "Tissue_
```



```
  rownames_to_column(var = "genes") %>%  
  filter(padj < 0.05) %>%  
  filter(abs(log2FoldChange) > 1) %>%  
  rename("CcoxCtsLog2FoldChange" = "log2FoldChange", "CcoxCtspadj" = "padj") %>%  
  select(genes, CcoxCtsLog2FoldChange, CcoxCtspadj)
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:  
##   Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.  
##   https://doi.org/10.1093/biostatistics/kxw041
```

```
# CRH
```

```
CRH = data.frame(lfcShrink(dds = Campy, contrast = list(c("SpeciesSmallerCrh.TissueSM", "Tissue_SM_vs_E
```



```
  rownames_to_column(var = "genes") %>%  
  filter(padj < 0.05) %>%  
  filter(abs(log2FoldChange) > 1) %>%  
  rename("CrhLog2FoldChange" = "log2FoldChange", "Crhpadj" = "padj") %>%  
  select(genes, CrhLog2FoldChange, Crhpadj)
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:  
##   Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.  
##   https://doi.org/10.1093/biostatistics/kxw041
```

```
#CTS
```

```
CTS = data.frame(lfcShrink(dds = Campy, contrast = list(c("SpeciesSmallerCts.TissueSM", "Tissue_SM_vs_EO")),  
  rownames_to_column(var = "genes") %>%  
  filter(padj < 0.05) %>%  
  filter(abs(log2FoldChange) > 1) %>%  
  rename("CtsLog2FoldChange" = "log2FoldChange", "Ctspadj" = "padj") %>%  
  select(genes, CtsLog2FoldChange, Ctspadj))
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:  
## Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.  
## https://doi.org/10.1093/biostatistics/kxw041
```

```
## Joining of result tables datasets
```

```
DEG_joined_full = inner_join(x = CCO, y = CCOxCRH, by = "genes")  
DEG_joined_full = inner_join(x = DEG_joined_full, y = CCOxCTS, by = "genes")  
DEG_joined_full = inner_join(x = DEG_joined_full, y = CRH, by = "genes")  
DEG_joined_full = inner_join(x = DEG_joined_full, y = CTS, by = "genes")
```

```
DEG_avg = DEG_joined_full %>%  
  select(contains("Log2FoldChange")) %>%  
  rowMeans()
```

```
DEG_padj = DEG_joined_full %>%  
  select(contains("padj")) %>%  
  rowMeans()
```

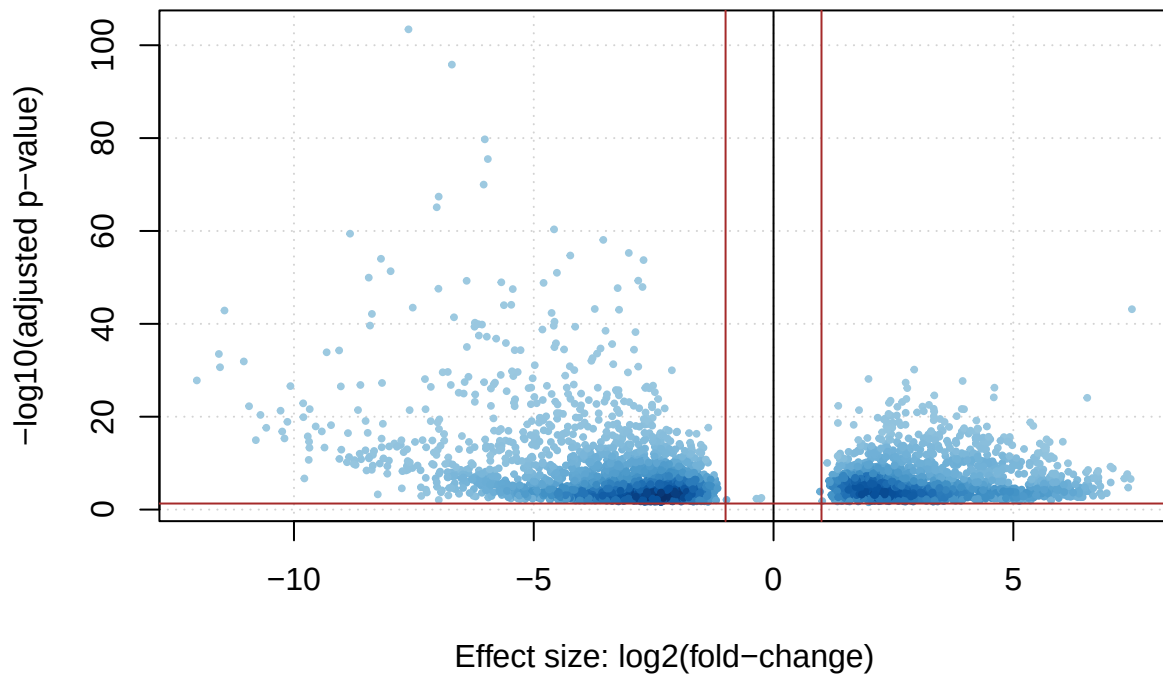
```
DEG_joined = DEG_joined_full %>%  
  mutate(overall_avg = DEG_avg) %>%  
  mutate(overall_padj = DEG_padj) %>%  
  select(genes, overall_avg, overall_padj)
```

```
alpha = 0.05
```

```
DEG_joined$log_padj = -log10(DEG_joined$overall_padj)
```

```
cols = densCols(DEG_joined$overall_avg, DEG_joined$log_padj)  
DEG_joined$pch = 19  
DEG_joined$pch[DEG_joined$pvalue == 0] = 6  
plot(DEG_joined$overall_avg,  
  DEG_joined$log_padj,  
  col=cols, panel.first=grid(),  
  main="Volcano Plot of SM Differential Expression with EO Baseline",  
  xlab="Effect size: log2(fold-change)",  
  ylab="-log10(adjusted p-value)",  
  pch=DEG_joined$pch, cex=0.4)  
abline(v=0)  
abline(v=c(-1,1), col="brown")  
abline(h=-log10(alpha), col="brown")
```

Volcano Plot of SM Differential Expression with EO Baseline



9. Visualize individual candidates

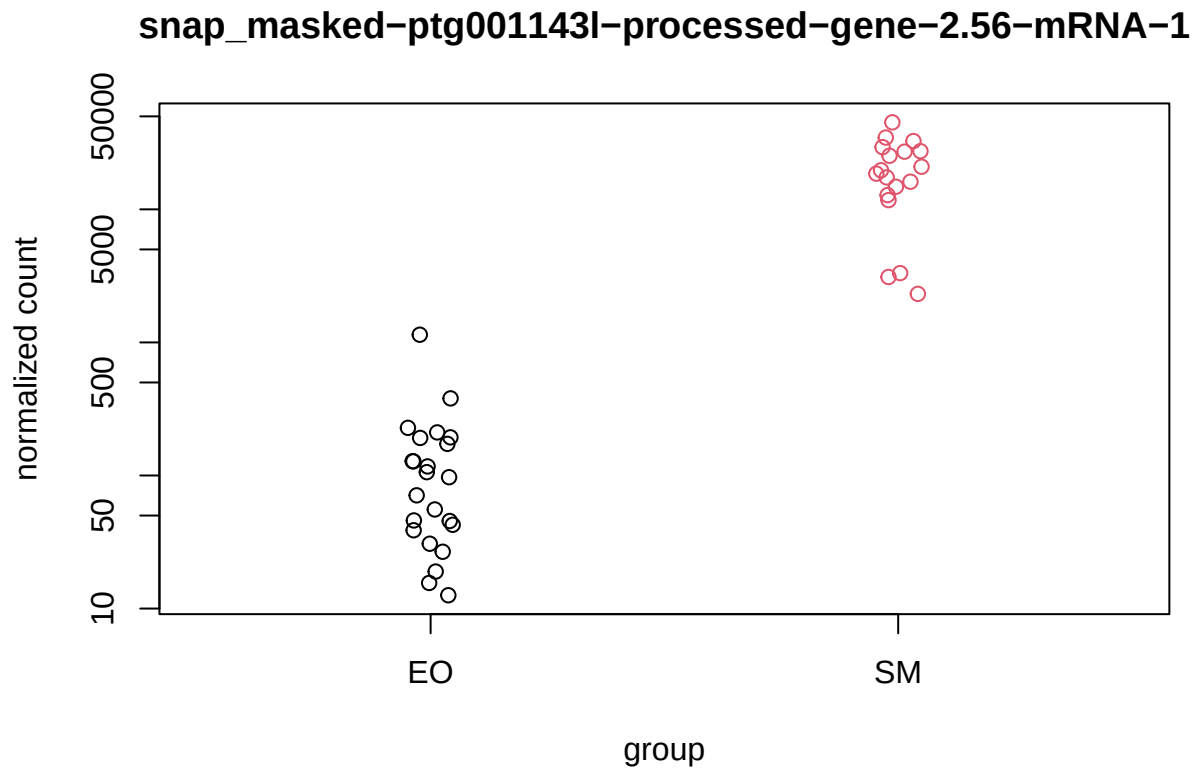
- Using the DEG dataframe from above, report out the top five candidates with the lowest average adjusted p-value.
- Using the DEG dataframe from above, visualize the normalized counts for the gene with the highest log2FC in the SM. Also, make a plot for the top candidate for the EO.

```
lowest_padj = DEG_joined %>%  
  select(genes, overall_padj) %>%  
  arrange(overall_padj) %>%  
  head(n = 5L)
```

```
highFC_gene = DEG_joined %>%  
  select(genes, overall_avg) %>%  
  arrange(overall_avg) %>%  
  tail(n = 1L)
```

```
Campy %>%
```

```
  plotCounts(gene = "snap_masked-ptg001143l-processed-gene-2.56-mRNA-1", intgroup = "Tissue", normalized = TRUE)
```

10. Using the DESeq LRT test, evaluate whether there is an association between gene expression and electric discharge signal duration

- Complete a LRT test using design formulas to determine whether there is an interaction between duration and electric discharge signal
 - You may need to reload in the data to DESeq2 and re-process it depending on how you specified your design formula.
 - Follow this resource to guide your approach
 - Use only the purebred species for this analysis (NO hybrids, i.e., do not include ccoxcts or ccoxrh in the analysis)
- Use degPatterns to sort genes into groups that correspond to EOD duration
 - From the output, identify groups that 1) have higher gene expression in the EO than SM and 2) have gene expression patterns that increase or decrease with EOD duration
 - Create a plot from degPlotCluster to showcase ONLY these selected groups (ex: a group of genes that has a higher abundance in longer compared to shorter duration electric signals)

YOUR CODE HERE

11. Based on the signal duration clustering, select 1 of the relevant clusters and complete a gene ontology analysis.

- Isolate a gene list for the cluster of interest
- Convert Campy gene names to D. rerio gene names using the orthology table derived from our ICA2 Genespace assignment (Danio_Campylomormyrus_1to1_orthologs_table.csv)

- Note: degPatterns transforms the Campy gene names, make sure to change them back so they match with the orthology table
- Complete a gene ontology analysis on the cluster of interest using the zebrafish gene names
 - Note: enrichGO and enrichKEGG are options, use this reference

#Include commands

Project 2 Part 5: Creating high quality figures for your report

Go back through your work from part 1-4 and identify the key pieces of information and plots for writing your project 2 report. Remake your chosen plots below, creating publication quality figures. Your project 2 report should have 4-6 figures, these can have multiple panels if you know how to achieve that in R. Each figure should add a different piece of information (i.e., do NOT include 3 dimensionality plots that show the same pattern).

Guidelines:

- Figures should have a title and axes that are easy to read
- Have an accessible color scheme
- Have scientifically relevant information highlighted (ex: color scheme denotes different species)
- When applicable, color schemes should be consistent across plots

1. Figure 1

#Make figure
#OR delete this box and link a figure that you made outside of R

2. Figure 2

#Make figure
#OR delete this box and link a figure that you made outside of R

3. Figure 3

#Make figure
#OR delete this box and link a figure that you made outside of R

4. Figure 4

#Make figure
#OR delete this box and link a figure that you made outside of R

5. Figure 5

#OPTIONAL analysis

5. Figure 6

#OPTIONAL analysis